

Predicting Kickstarter Campaign Success

Final Project Report

Data Science Capstone by Fahad Memon

1. Business Problem and Project Objective

Kickstarter creators need to make important decisions before launch, like how much money to ask for, how long the campaign should run, and when it should launch. A prediction model can help show the risk earlier, so the strategy can be adjusted. The objective of this project was to build a practical classifier that predicts campaign success from information available at launch and to turn the results into recommendations a creator or platform team can use.

2. Data and Preparation

The raw project data came from Kickstarter campaign records. During data wrangling, project states that were not final, such as `live` and `undefined`, were removed, malformed country entries were excluded, and campaign duration was calculated from the launch and deadline timestamps. After cleaning, the working dataset contained **372,066 rows** and 15 variables.

For modeling, the target was changed to a binary outcome where successful = 1 and all other final outcomes = 0. To keep the model useful before launch, leakage variables from after launch, such as pledged amount, real pledged amount, and backers, were removed. I also created calendar features from the launch timestamp, and categorical variables were one-hot encoded.

Preparation step	What was done
Target construction	Created state_binary where successful = 1 and unsuccessful = 0.
Leakage prevention	Dropped pledged, usd_pledged_real, backers, state, ID, and name.
Feature engineering	Derived duration, launch_year, launch_month, and launch_weekday.
Encoding and scaling	One-hot encoded category, main_category, currency, and country; standardized numeric fields.
Train/test split	Used a stratified 80/20 split, yielding 297,649 training rows and 74,413 test rows.

3. Exploratory Findings

The cleaned dataset was imbalanced, but it was still large: 133,851 campaigns (36.0%) were successful, while 238,215 (64.0%) were unsuccessful. Three patterns were clear in the exploratory

analysis. First, successful projects usually asked for less money. The median goal for successful campaigns was about \$3,840, compared with \$7,702 for unsuccessful campaigns. Second, success rates changed a lot by category. Third, timing also mattered, but the timing effect was smaller than the goal and category effects.

Funding goal and outcome

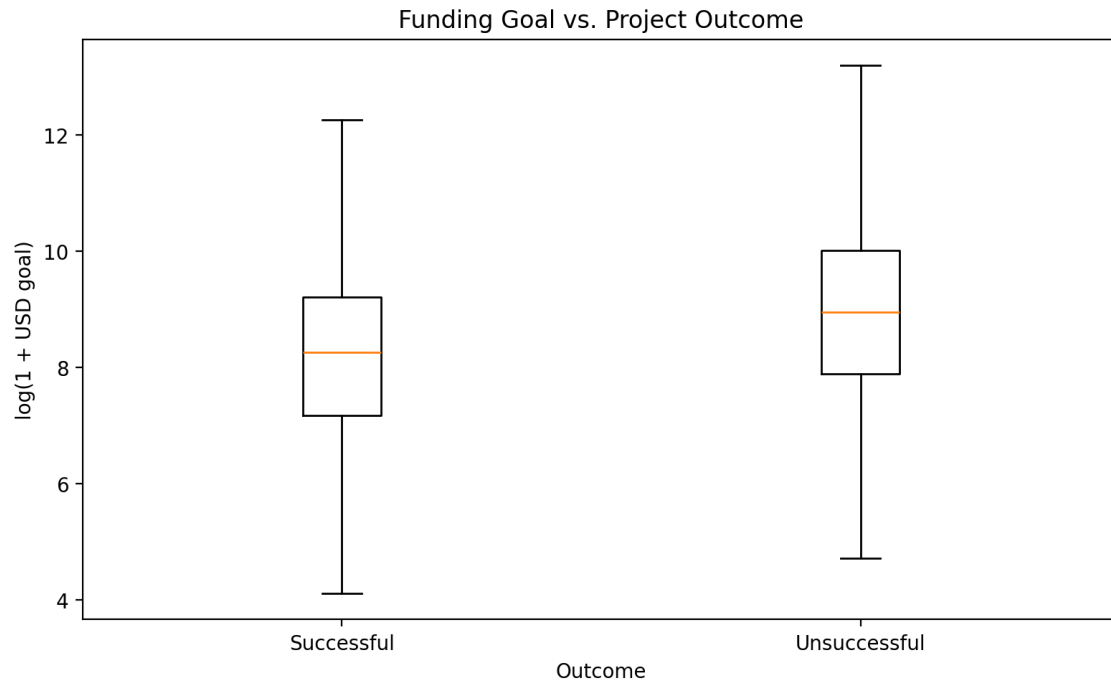


Figure 1. Successful projects usually had lower funding goals, even after using a log transform to reduce extreme values.

The difference in goal size was not only easy to see in the figure, but it was also strong in the exploratory notebook. In practical terms, campaigns with larger asks seem harder to fund, so setting a realistic goal should be treated as one of the most important campaign decisions.

Category differences

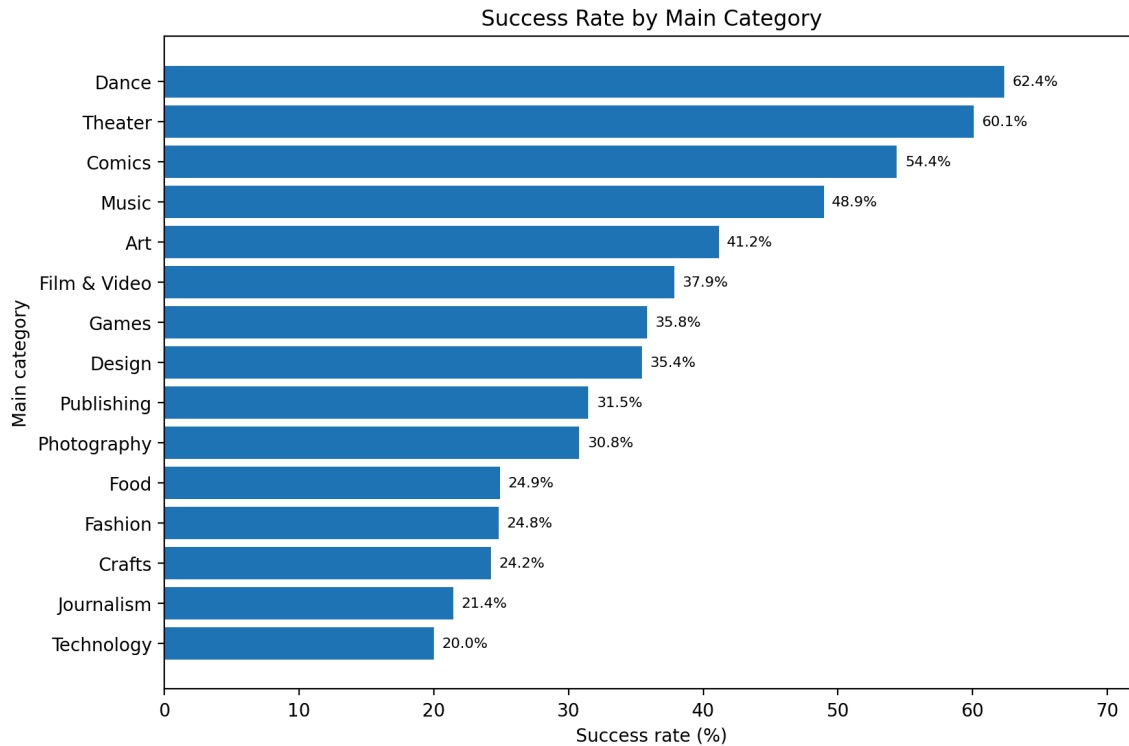


Figure 2. Success rates were very different across main categories, from 20.0% in Technology to 62.4% in Dance.

The strongest categories in this dataset were Dance, Theater, and Comics, while Technology, Journalism, and Crafts were the weakest. This means the same launch strategy may not work equally well for every project type.

Timing effects

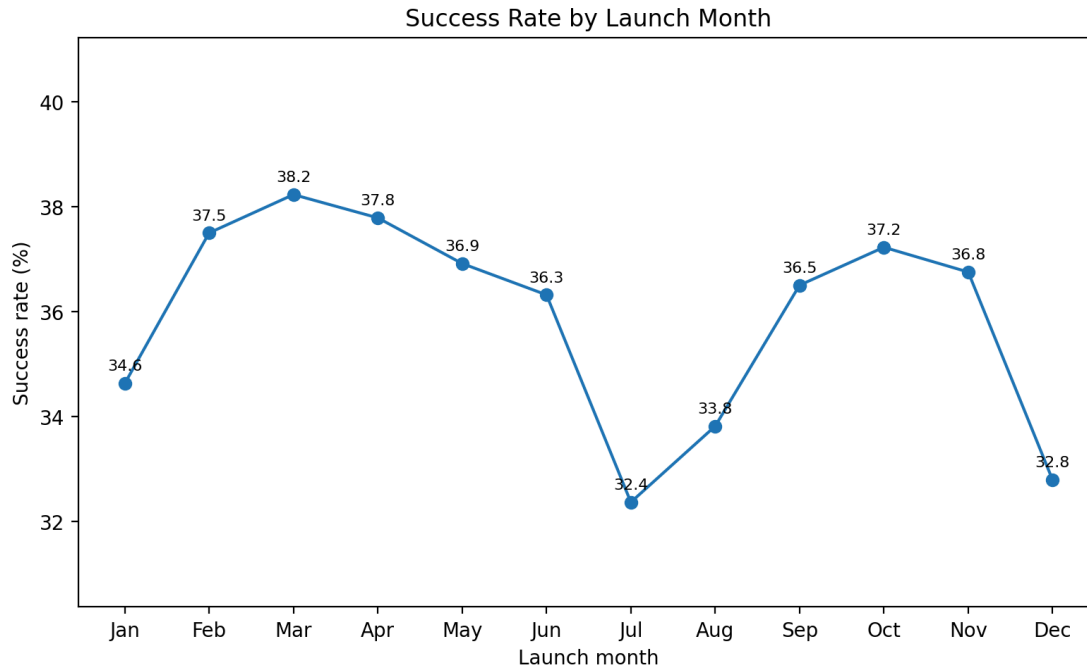


Figure 3. Launch timing had a modest but visible relation with success rate; March performed best (38.2%), while July was the weakest (32.4%).

The weekday pattern was smaller but showed a similar direction: Tuesday launches performed best at 37.8%, while Saturday was the lowest at 33.3%. These timing effects should be seen as secondary factors, not the main drivers, but they can still help with launch planning.

4. Modeling Approach

Four supervised classifiers were compared: logistic regression, a decision tree using the Gini criterion, a decision tree using the entropy criterion, and a random forest tuned with GridSearchCV. Because the classes were imbalanced, F1 score was used as the main metric for model selection. This is useful because it balances precision and recall, instead of rewarding a model that mostly predicts the majority class.

The random forest tuning grid searched $n_estimators = \{100, 200\}$, $max_depth = \{None, 10, 20\}$, $min_samples_split = \{2, 5\}$, and $min_samples_leaf = \{1, 2\}$ with 3-fold cross-validation.

5. Model Results

A dummy baseline that always predicts the majority class would get about 0.640 accuracy on the test set, but its F1 score would be 0.000 because it never predicts successful projects. All candidate models improved on this baseline for F1, and the tuned random forest was the best overall performer.

Model	Accuracy	Precision	Recall	F1 score
Logistic Regression	0.689	0.607	0.388	0.473

Decision Tree (Gini)	0.687	0.602	0.382	0.467
Decision Tree (Entropy)	0.686	0.607	0.365	0.456
Random Forest (Tuned)	0.696	0.601	0.457	0.520

The selected random forest reached 0.696 accuracy, 0.601 precision, 0.457 recall, and 0.520 F1 on the holdout set. Its confusion matrix ([[39,522, 8,121], [14,526, 12,244]]) shows that the model still misses a meaningful number of successful campaigns, but it identifies much more positive cases than the baseline and does this with better balance than the benchmark models.

What drove the final model

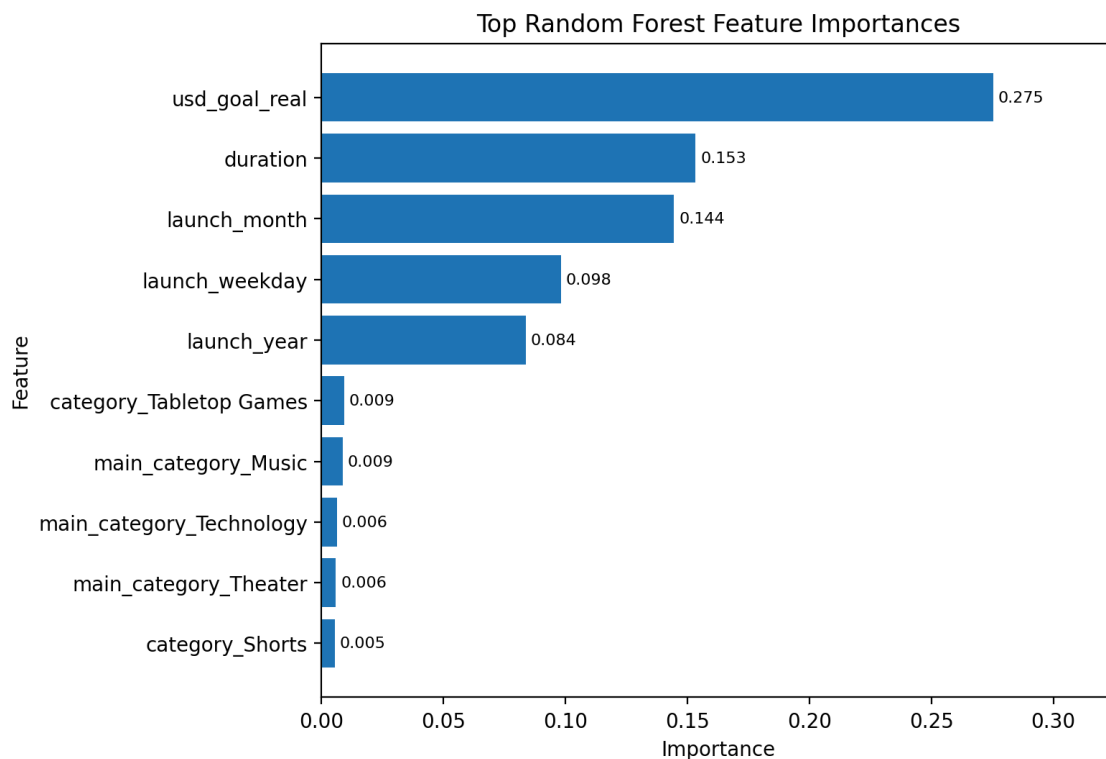


Figure 4. The final random forest relied most heavily on goal size, duration, and launch timing variables, followed by selected category indicators.

Feature importance supported the exploratory story. Goal size was the strongest predictor by a wide margin, followed by campaign duration and launch timing features. Category indicators also added signal, which supports giving category-specific launch guidance instead of using one strategy for all projects.

6. Recommendations

1. **Set more realistic funding goals.** Goal amount was the strongest signal in both the exploratory analysis and the final model. Creators should compare their funding ask with similar campaigns and check if the target looks realistically reachable.
2. **Keep the campaign length controlled.** Duration was the second most important feature in the final model. Successful projects were more common among shorter campaigns, so using around 30 days by default seems reasonable unless there is a strong reason to go longer.
3. **Use category-based launch planning and timing.** Category performance was very different, and small timing effects also appeared by month and weekday. Teams in lower-performing areas like Technology or Journalism should use more careful expectations and stronger pre-launch preparation.

7. Limitations and Further Research

This model was intentionally limited to structured features available at or before launch. That makes the predictions more actionable, but it also leaves out other possibly important information, such as project descriptions, images, videos, reward tiers, and creator history. The analysis also grouped all unsuccessful final states together, which makes the business question simpler but may hide meaningful differences between failed, canceled, and suspended projects.

Future work should add text-based features from titles and descriptions, test boosted tree methods such as XGBoost or LightGBM, calibrate predicted probabilities, and evaluate threshold tuning based on stakeholder priorities. A time-based validation design would also help show how well the model generalizes across different Kickstarter periods.

8. Conclusion

This project shows that Kickstarter outcomes can be predicted better than a majority-class baseline by using only launch-time campaign information. The best-performing model was a tuned random forest, and its strongest signals were intuitive: realistic goals, controlled campaign length, and category-aware planning matter. The model is not accurate enough to replace human judgment, but it can still be useful as a decision-support tool for screening campaigns and showing where creators may need to revise their strategy before launch.